DIGITAL DESIGN AND COMPUTER ARCHITECTURE

Design of a 4-Bit ALU Using Multiplexer

AND, OR and Set-Less-Than (SLT) Operations

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PROJECT OVERVIEW

A 4-bit ALU combines three operations under one control interface

Supported operations

An Arithmetic Logic Unit is a CPU component that performs arithmetic and logical operations on binary data.

- AND: bitwise logic on two 4-bit inputs.
- OR: bitwise logic on two 4-bit inputs.
- SLT: unsigned comparison of A and B.

A multiplexer selects the required result using two control signals, S1 and S0.

Project objectives

- Design a 4-bit ALU using basic digital logic.
- Implement AND and OR for two 4-bit inputs.
- Compare unsigned 4-bit values using SLT.
- Verify the circuit in Logisim or Logisim Evolution.

CORE OPERATIONS

AND and OR are bitwise; SLT compares the full unsigned values

Each operation produces a 4-bit candidate result. The multiplexer later chooses which candidate reaches Y[3:0].

AND

Each output bit is 1 only when both corresponding input bits are 1.

Y[i] = A[i] AND B[i]

OR

Each output bit is 1 when at least one corresponding input bit is 1.

Y[i] = A[i] OR B[i]

Unsigned SLT

If A < B, output **0001**; otherwise output **0000**. SLT is a magnitude comparison, not a bitwise operation.

Operation selection

S1	S0	Selected operation	Output
0	0	AND	A AND B
0	1	OR	A OR B
1	0	SLT	0001 if A < B, otherwise 0000
1	1	Reserved	Unspecified; do not use

LOGISIM IMPLEMENTATION

Logisim implementation uses parallel logic and a 4-bit multiplexer

All candidate results are computed before the selected result is displayed.

Required components

- Two 4-bit inputs: A[3:0] and B[3:0]
- Four 2-input AND gates and four 2-input OR gates
- One unsigned 4-bit magnitude comparator
- Logic that converts A < B into 0001 or 0000
- One 4-bit-wide 4-to-1 multiplexer
- Two select signals, S1 and S0, and a 4-bit output Y[3:0]

Working sequence

1. Enter two 4-bit binary values.
2. Compute AND and OR results in parallel.
3. Compare A and B as unsigned values.
4. Form the four-bit SLT result.
5. Select the result with S1S0 and display Y[3:0].

Implementation placeholder: [INSERT GENUINE LOGISIM CIRCUIT SCREENSHOT HERE]

Reference only; complete implementation and verification in Logisim.

Expected outputs are ready for verification in Logisim

For each test, S1S0 = 00 selects AND, 01 selects OR, 10 selects SLT, and 11 has no defined functional result.

Control selection

S1S0	Operation
00	AND
01	OR
10	SLT
11	Reserved/Unused

Expected test cases — pending Logisim verification

Test	A	B	AND	OR	SLT
1	0101 (5)	0011 (3)	0001	0111	0000
2	0010 (2)	0110 (6)	0010	0110	0001
3	1111 (15)	1111 (15)	1111	1111	0000
4	0000 (0)	0001 (1)	0000	0001	0001

Simulation placeholder: [INSERT GENUINE LOGISIM SIMULATION SCREENSHOT HERE]

CONCLUSION · FUTURE SCOPE

The design demonstrates combinational logic and controlled result selection

Conclusion

- The conceptual 4-bit ALU supports AND, OR, and unsigned SLT.
- Two control signals select the required result.
- Logisim can be used for implementation and verification.

Applications

- CPU and processor datapaths
- Digital electronics and computer architecture education
- Embedded-system learning

Future improvements

- Addition and subtraction
- XOR and NOT operations
- 8-bit or 16-bit inputs
- Integration into a simple processor datapath

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Thank you

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